

Shared cointegration: Multiple time series sharing a cointegrating relationship

Title TBD

TEAM B4

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Team Members

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September 30, 2025



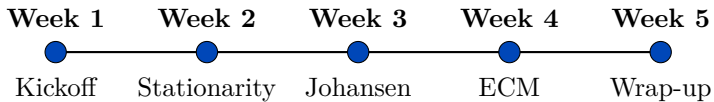
① Timeline

② References

1 Timeline

2 References

Timeline



Week 1

Period: (Oct 7 – Oct 13)

- Team discussion & work distribution (who handles dataset search, coding, slides, etc.).
- Finalize dataset (e.g., macroeconomic/financial variables).
- Everyone revises basics: stationarity, ADF/PP tests, cointegration concept.

Week 2

Period: (Oct 14 – Oct 20)

- Perform unit root tests (ADF/PP) on chosen variables.
- Share results among team & discuss division of analysis.
- Start applying Engle–Granger 2-step test for initial cointegration check.

Week 3

Period: (Oct 21 – Oct 27)

- Run Johansen cointegration test for multiple series.
- Identify cointegrating vectors & interpret results.
- Team sync-up: assign one subgroup to results interpretation and one to visuals/plots.

Week 4

Period: (Oct 28 – Nov 3)

- Develop Error Correction Model (ECM) to show short-run vs long-run dynamics.
- Validate with diagnostic checks and finalize output tables/graphs.
- Start making presentation slides (intro, methodology, results, discussion).

Week 5

Period: (Nov 4 – Nov 7)

- Team discussion & rehearsal: combine slides from all members.
- Final check: clarity, timing, flow of presentation.
- Submit final slides/report if required.
- Practice dry run before presentation day.

① Timeline

② References

References

- [1] “Cointegration tests.” <https://www.geeksforgeeks.org/data-science/cointegration-tests/>, 2025.
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- [2] “Cointegration.” <https://en.wikipedia.org/wiki/Cointegration>, 2025.
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- [3] “Cointegration.” <https://corporatefinanceinstitute.com/resources/data-science/cointegration/>, n.d.
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- [4] R. Susmel, “Econometrics lecture ec2-7.” <https://www.bauer.uh.edu/rsusmel/phd/ec2-7.pdf>.
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- [5] B. Sorensen, “Cointegration: Theory and tests.” <https://www.uh.edu/~bsorensen/coint.pdf>.
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Thank You